$$= \int_0^{\frac{\pi}{4}} \log\left[\frac{2}{1+\tan x}\right] dx = \int_0^{\frac{\pi}{4}} \left[\log 2 - \log(1+\tan x)\right] dx$$

$$= \log 2 \int_0^{\frac{\pi}{4}} dx - \int_0^{\frac{\pi}{4}} \log(1+\tan x) dx$$

$$= \frac{\pi}{4}\log 2 - I$$

So, we get $2I = \frac{\pi}{4}\log 2$. Hence, we get $I = \frac{\pi}{8}\log 2$.

Example 9.28

Show that $\int_0^1 \left(\tan^{-1} x + \tan^{-1}(1-x)\right) dx = \frac{\pi}{2} - \log_e 2$.

Solution

$$I = \int_0^1 \left(\tan^{-1} x + \tan^{-1}(1-x)\right) dx$$

$$= \int_0^1 \tan^{-1} x \, dx + \int_0^1 \tan^{-1}(1-x) \, dx$$

$$= \int_0^1 \tan^{-1} x \, dx + \int_0^1 \tan^{-1}(1-(1-x)) \, dx \text{, since } \int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

$$= \int_0^1 \tan^{-1} x \, dx + \int_0^1 \tan^{-1} x \, dx$$

$$= 2\int_0^1 \tan^{-1} x \, dx$$

$$= \left[2\int u dv\right]_0^1 \text{, where } u = \tan^{-1} x \text{ and } dv = dx$$

$$= 2\left[uv - \int v du\right]_0^1 \text{, applying integration by parts}$$

$$= 2\left(x\tan^{-1} x - \int x \frac{dx}{1+x^2}\right)_0^1 = 2\left(x\tan^{-1} x - \frac{1}{2}\log\left(1+x^2\right)\right)_0^1 = \frac{\pi}{2} - \log 2$$

Example 9.29

Evaluate $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}} dx$.

Solution

Let us put $I = \int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}} dx$ ... (1)

Applying the formula $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$, we get

$$I = \int_2^3 \frac{\sqrt{(2+3-x)}}{\sqrt{5-(2+3-x)}+\sqrt{(2+3-x)}} dx = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{x}+\sqrt{5-x}} dx \quad \text{... (2)}$$

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